
IK 517: Philosophy of Science

Course Code: IK 517

Course Name: Philosophy of Science

L-T-P-C: 3-0-0-3

Prerequisites: -

Students intended for:

Elective or Core: Elective

Mutual Exclusion: -

Approval: 59th BoA

Course contents

- **Foundations of the Philosophy of Science** [4 Lectures]
 - Introduction to the Philosophy of Science
 - Historical Background and Key Concepts
 - The Role of Philosophy in Science
- **Laws, Explanation, and Prediction** [4 Lectures]
 - The Value of Laws in Science
 - Explanation and Prediction in Scientific Theories
 - Case Studies in Scientific Explanation
- **Induction and Probability** [4 Lectures]
 - Induction and Statistical Probability
 - Induction and Logical Probability
 - Challenges to Inductive Reasoning
- **Measurement and Quantitative Language** [4 Lectures]
 - The Experimental Method
 - Quantitative Concepts in Science
 - Extensive Magnitudes and Measurement
- **Space, Time, and Geometry** [4 Lectures]
 - Euclidean and Non-Euclidean Geometries
 - Space in Relativity Theory
 - Philosophical Implications of Space and Time
- **Causality and Determinism** [4 Lectures]
 - Causality in Science
 - Determinism and Free Will

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- Case Studies in Causality and Determinism
 - **Theories and Nonobservables** [5 Lectures]
 - Theories and Correspondence Rules
 - The Ramsey Sentence
 - Analyticity in Scientific Theories
 - **Statistical Laws and Quantum Mechanics** [5 Lectures]
 - Statistical Laws and Indeterminism
 - The Philosophy of Quantum Physics
 - Case Studies in Quantum Mechanics
 - **Ethics, Technology, and Social Dimensions of Science** [4 Lectures]
 - Ethics in Science
 - The Role of Technology in Science
 - Social Dimensions of Science
 - **Philosophy of Scientific Revolutions and Contemporary Issues** [4 Lectures]
 - Philosophy of Scientific Revolutions
 - Contemporary Issues in Philosophy of Science
 - Review and Synthesis

Textbooks and References

1. Rudolf Carnap, **An Introduction to the Philosophy of Science**, Primary Text.
2. James Ladyman, **Understanding Philosophy of Science**, Supplementary Text.